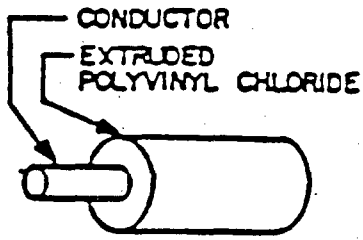
INCH-POUND
MIL-W-16878/1B(NAVY)
11 September 1992
SUPERSEDING
MIL-W-16878/1A(NAVY)
10 August 1981

MILITARY SPECIFICATION SHEET

WIRE, ELECTRICAL, POLYVINYL CHLORIDE (PVC) INSULATED, 105°C, 600 VOLTS (NOT FOR NAVY SHIPBOARD USE)

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-W-16878.



(NOT FOR NAVY SHIPBOARD USE)

FIGURE 1. Construction.

AMSC N/A FSC 6145
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:25Z
Check the source to verify that this is the current version before use.

TABLE I. Construction details.

(See footnotes at end of table.)

PIN 4/	Wire size	Stranding	Conductor		Conductor diameter (nominal) (inch)	Finished wire diameter (inch)	
			Material 2/ 3/	Coating		Min	Max
M16878/1-BAA*	32	1 X 32	Copper	Tin	0.0080	0.024	0.030
M16878/1-DAA*	32	1 X 32	H.S.C.A.	Silver	.0080	.024	.030
M16878/1-CAA*	32	1 X 32	C.C. steel	Tin	.0080	.024	.030
M16878/1-BAB*	32	7 X 40	Copper	Tin	.010	.026	.032
M16878/1-DAB*	32	7 X 40	H.S.C.A.	Silver	.010	.026	.032
M16878/1-BBA*	30	1 X 30	Copper	Tin	.0100	.026	.032
M16878/1-DBA*	30	1 X 30	H.S.C.A.	Silver	.0100	.026	.032
M16878/1-CBA*	30	1 X 30	C.C. steel	Tin	.0100	.026	.032
M16878/1-BBB*	30	7 X 38	Copper	Tin	.012	.028	.034
M16878/1-DBB*	30	7 X 38	H.S.C.A.	Silver	.012	.028	.034
M16878/1-BCA*	28	1 X 28	Copper	Tin	.0126	.029	.035
M16878/1-DCA*	28	1 X 28	H.S.C.A.	Silver	.0126	.029	.035
M16878/1-CCA*	28	1 X 28	C.C. steel	Tin	.0126	.029	.035
M16878/1-BCB*	28	7 X 36	Copper	Tin	.015	.031	.037
M16878/1-DCB*	28	7 X 36	H.S.C.A.	Silver	.015	.031	.037
M16878/1-BDA*	26	1 X 26	Copper	Tin	.0159	.032	.038
M16878/1-DDA*	26	1 X 26	H.S.C.A.	Silver	.0159	.032	.038
M16878/1-CDA*	26	1 X 26	C.C. steel	Tin	.0159	.032	.038
M16878/1-BDB*	26	7 X 34	Copper	Tin	.019	.035	.041
M16878/1-DDB*	26	7 X 34	H.S.C.A.	Silver	.019	.035	.041
M16878/1-BDE*	26	19 X 38	Copper	Tin	.021	.035	.041
M16878/1-DDE*	26	19 X 38	H.S.C.A.	Silver	.021	.035	.041
M16878/1-BEA*	24	1 X 24	Copper	Tin	.0201	.036	.043
M16878/1-DEA*	24	1 X 24	H.S.C.A.	Silver	.0201	.036	.043
M16878/1-CEA*	24	1 X 24	C.C. steel	Tin	.0201	.036	.043
M16878/1-BEB*	24	7 X 32	Copper	Tin	.024	.040	.047
M16878/1-DEB*	24	7 X 32	H.S.C.A.	Silver	.024	.040	.047
M16878/1-BEE*	24	19 X 36	Copper	Tin	.026	.040	.047
M16878/1-DEE*	24	19 X 36	H.S.C.A.	Silver	.026	.040	.047
M16878/1-BFA*	22	1 X 22	Copper	Tin	.0253	.041	.048
M16878/1-DFA*	22	1 X 22	H.S.C.A.	Silver	.0253	.041	.048
M16878/1-CFA*	1/ 22	1 X 22	C.C. steel	Tin	.0253	.041	.048
M16878/1-BFB*	22	7 X 30	Copper	Tin	.030	.046	.053
M16878/1-DFB*	22	7 X 30	H.S.C.A.	Silver	.030	.046	.053
M16878/1-BFE*	22	19 X 34	Copper	Tin	.032	.046	.053
M16878/1-DFE*	22	19 X 34	H.S.C.A.	Silver	.032	.046	.053
M16878/1-BGA*	20	1 X 20	Copper	Tin	.032	.048	.055
M16878/1-DGA*	20	1 X 20	H.S.C.A.	Silver	.032	.048	.055
M16878/1-BGB*	20	7 X 28	Copper	Tin	.038	.054	.061
M16878/1-DGB*	20	7 X 28	H.S.C.A.	Silver	.038	.054	.061
M16878/1-BGE*	20	19 X 32	Copper	Tin	.041	.054	.061

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:25Z
Check the source to verify that this is the current version before use.

PIN 4/	Wire size	Strand-ing	Conductor		Conductor diameter (nominal) (inch)	Finished wire diameter (inch)	
			Material 2/ 3/	Coating		Min	Max
M16878/1-BHA*	18	1 X 18	Copper	Tin	0.0403	0.056	0.063
M16878/1-BHB*	18	7 X 26	Copper	Tin	.049	.065	.072
M16878/1-BHE*	18	19 X 30	Copper	Tin	.051	.065	.072
M16878/1-BJA*	16	1 X 16	Copper	Tin	.0508	.067	.075
M16878/1-BJE*	16	19 X 29	Copper	Tin	.059	.075	.083
M16878/1-BKA*	14	1 X 14	Copper	Tin	.0641	.080	.088
M16878/1-BKE*	14	19 X 27	Copper	Tin	.072	.088	.096

1/ Inactive for new design.
2/ H.S.C.A. stands for high-strength copper alloy.
3/ C.C. stands for copper-clad.
4/ PIN stands for part or identifying number.

ADDITIONAL REQUIREMENTS:

Visual and mechanical examination: Required.
Spark test: 3.4 Kilovolts (kV)
Impulse dielectric test: 8.0 kV.
Dielectric withstanding voltage: 2.0 kV.
Insulation resistance: $IR = K \log_{10} D/d$

Where: IR = Minimum insulation resistance in megohms per 1000 feet at 20°C.
K = 2000.
D = Maximum average diameter of finished wire.
d = Conductor diameter.

Cold bend: Condition 4 hours at minus 54 ± 1°C (see table II).

TABLE II. Cold bend mandrel sizes.

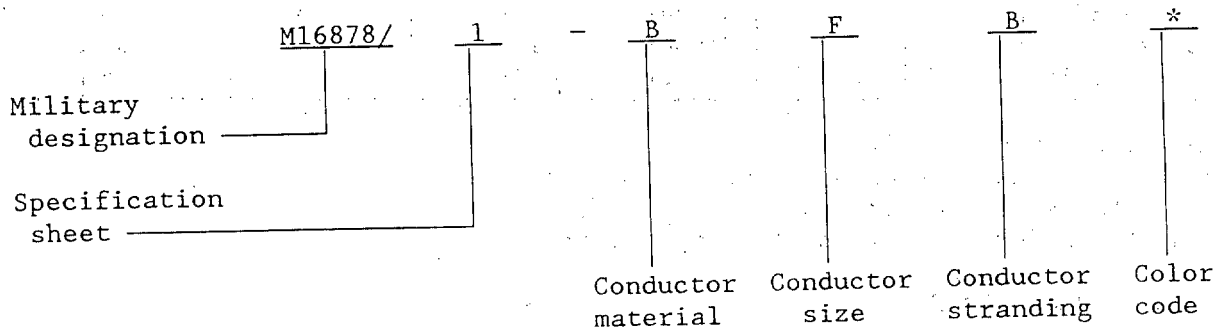
Wire size	Cold bend mandrel (maximum diameter in inches)
32 through 26	1
24 through 14	2

Concentricity: 70 percent (minimum).
Surface resistance: Not required.
Wrap back: Not required.
Heat resistance: Condition at 150°C.
Flammability: Meets requirements of UL VW1 Flame Test.
Toxicity: Meets requirements of concentration levels specified in MIL-E-917.
Shrinkage: Required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:25Z
Check the source to verify that this is the current version before use.

Heating aging: 25 percent change (maximum) in 96 hours at 136°C.
Insulation tensile strength: 1800 pounds force per square inch (minimum).
Insulation elongation: 100 percent (minimum).
Marking and stripe durability: Required.
Fungus resistance: Required.
Maximum dc resistance of finished wire: See table I of MIL-W-16878.
Identification of product: Required for wire sizes 22 AWG and larger.

PIN example (see MIL-W-16878):



Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6145-N330-01)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:25Z
Check the source to verify that this is the current version before use.